

$$DA_2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \text{ (By defn. of } Ax)$$

$$= \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 6 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 20 \end{bmatrix} \text{ (By defn. of scalar multiple)}$$

$$= \begin{bmatrix} 2 \\ 6 \\ 20 \end{bmatrix} \text{ (By defn. of vector sum)}$$

The second column of DA is a linear combination of the columns of D where the weights are the corresponding entries of the 2nd column of A .

$$DA_3 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \text{ (By defn. of } Ax)$$

$$= \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 9 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 25 \end{bmatrix} \text{ (By defn. of scalar multiple)} = \begin{bmatrix} 2 \\ 9 \\ 25 \end{bmatrix} \text{ (By defn. of vector sum)}$$

$DA = \begin{bmatrix} 2 & 2 & 2 \\ 3 & 6 & 9 \\ 5 & 20 & 25 \end{bmatrix}$ The third column of DA is a linear combination of the columns of D where the weights are the corresponding entries of the 3rd column of A .

$$\text{Let } B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \cdot AB = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} Ab_1 & Ab_2 & Ab_3 \end{bmatrix} \text{ (By defn. of matrix multiplication)}$$

$$Ab_1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = b_{11} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + b_{21} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + b_{31} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \text{ (By defn. of } Ax)$$

$$= \begin{bmatrix} b_{11} \\ b_{11} \\ b_{11} \end{bmatrix} + \begin{bmatrix} b_{21} \\ 2b_{21} \\ 4b_{21} \end{bmatrix} + \begin{bmatrix} b_{31} \\ 3b_{31} \\ 5b_{31} \end{bmatrix} \text{ (By defn. of scalar multiple)}$$

$$= \begin{bmatrix} b_{11} + b_{21} + b_{31} \\ b_{11} + 2b_{21} + 3b_{31} \\ b_{11} + 4b_{21} + 5b_{31} \end{bmatrix} \text{ (By defn. of vector sum)}$$

$$Ab_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = b_{12} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + b_{22} \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + b_{32} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \text{ (By defn. of } Ax)$$